

Module 14: Macroevolution — Questions

Quick Check

- 1. Under the Biological Species Concept, a species is defined primarily by:** A) Its physical appearance B) Its ability to successfully interbreed and produce fertile offspring C) Its geographic location D) Its position on a cladogram
 - 2. Two species of closely related frogs live in the same pond. Species A mates exclusively in early March, while Species B mates exclusively in late May. This is an example of:** A) Mechanical Isolation B) Habitat Isolation C) Temporal Isolation D) Reduced Hybrid Fertility
 - 3. A male lion and a female tiger mate in captivity to produce a "liger." Ligers are massive and generally healthy animals, but they are completely sterile and cannot have offspring. This is an example of a:** A) Prezygotic barrier B) Postzygotic barrier C) Gametic barrier D) Sympatric barrier
 - 4. A deep canyon forms over millions of years, splitting a population of mice. The environments on either side of the canyon are slightly different, and the mice slowly adapt to their separate cliffs. If the mice are later reunited, they can no longer interbreed. This is an example of:** A) Allopatric speciation B) Sympatric speciation C) Artificial selection D) The founder effect
 - 5. True or False:** Sympatric speciation requires a physical geographic barrier (like a mountain or ocean) to split the population. - ☐ True - ☐ False
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Reflection

- 6. Explain why the Biological Species Concept cannot be used to classify a newly discovered fossil of a dinosaur.**

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7. Differentiate between prezygotic and postzygotic barriers. Provide one unique real-world example of each.

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8. Imagine a massive hurricane blows ten birds from an island to a new, isolated island miles out to sea. Explain how this single event involves BOTH microevolution (Module 13) and macroevolution (Module 14) over the next two million years.

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